

SUKADEV YARLAPALLI

AI / ML Engineer — LLMs, Agentic AI & MLOps

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PROFESSIONAL SUMMARY

Builder-minded AI/ML engineer focused on shipping LLM-powered products end-to-end. Hands-on with LangChain, agentic workflows (planning, tool use, multi-step reasoning), recommendation systems, and MLOps (CI/CD, deployment, monitoring). Comfortable taking an idea from zero to a shipped product, iterating fast, and owning outcomes in a startup environment.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JSON, XML
- **LLMs & Agents:** LangChain, LangGraph, OpenAI / Gemini / Groq APIs, RAG, Function/Tool Calling, Prompt Engineering, Vector DBs (FAISS, Chroma, Pinecone)
- **Agentic AI:** ReAct, Planner–Executor, Multi-step Reasoning, Tool Use, Memory, Autonomous Workflows
- **ML / Data:** Scikit-Learn, Pandas, NumPy, NLP, OCR, Recommendation Systems (Collaborative + Content-based, Hybrid), Feature Engineering
- **MLOps & Deployment:** Docker, GitHub Actions (CI/CD), FastAPI, Streamlit, Hugging Face Spaces, Model Monitoring, Logging
- **Automation & BI:** Power Automate, PowerApps, Power BI, Databricks, REST APIs

PROJECTS

Agentic Research Assistant (LLM + LangChain Agents)

Python, LangChain, LangGraph, OpenAI, FAISS, FastAPI, Docker

- Built a multi-step agentic AI that takes a single prompt, plans sub-tasks, picks tools (web search, RAG over PDFs, Python REPL, SQL), and ships a finished research brief without human hand-holding.
- Implemented a Planner–Executor loop with LangGraph + ReAct reasoning, tool-calling, and short/long-term memory backed by FAISS vector store.
- Wrapped the agent in a FastAPI service, containerized with Docker, and deployed with GitHub Actions CI/CD — average task completion in under 40 seconds end-to-end.

Personalized Content Recommendation Engine

Python, Scikit-Learn, Implicit, FastAPI, Streamlit

- Designed a hybrid recommendation engine combining collaborative filtering (ALS) and content-based embeddings to predict what users actually want next.
- Boosted offline top-10 recall by 28% vs. a popularity baseline on a 100K+ interaction dataset; added a cold-start fallback using LLM-generated item embeddings.
- Exposed real-time recommendations through a FastAPI endpoint with sub-100ms latency and an interactive Streamlit demo.

LLM-Powered Workflow Automation Agent

Python, LangChain, OpenAI Functions, REST APIs, Power Automate

- Built an agent that converts plain-English requests ("summarize yesterday's tickets and email the team") into executed multi-step work across Gmail, Sheets, and SharePoint.
- Used OpenAI function-calling + LangChain tools to safely orchestrate APIs with retries, validation, and audit logs — eliminated 15+ hours/week of manual ops work.
- Added guardrails: structured output schemas, action confirmation, and per-tool rate limiting.

MLOps Pipeline for Continuous Model Delivery

Python, Docker, GitHub Actions, FastAPI, Hugging Face, Prometheus-style logging

- Stood up a full MLOps pipeline: data validation → training → evaluation → containerized deploy → live monitoring, triggered automatically on every git push.
- Used GitHub Actions for CI/CD, Docker for reproducible builds, and a lightweight monitoring layer to track drift, latency, and error rates in production.
- Cut model release cycle from days to under 15 minutes, enabling ship-fast / iterate-fast experimentation.

Intelligent Document Reader (NLP + OCR)

Python, Tesseract OCR, NLP, Regex, LangChain

- Developed an OCR + NLP pipeline that extracts, classifies, and summarizes scanned invoices and PDFs with 94% extraction accuracy.
- Layered an LLM on top to answer natural-language questions over the extracted content (RAG-style).
- Reduced document processing time from ~6 minutes to under 30 seconds per file.

EDUCATION

Imarticus Learning, Hyderabad — Post-Graduation in Data Science & Analytics (Advanced ML Track), 2025–2026

Oxford Degree College, Sri Venkateswara University — B.Sc. (Maths, Physics, Chemistry), 2022–2025 | CGPA: 6.5

CERTIFICATIONS

- Post-Graduation Certification in Data Science & Analytics with Advanced ML Track — Imarticus Learning
- Python for Data Science & Machine Learning — Self-paced, project-based learning

ACHIEVEMENTS

- Shipped 5+ end-to-end AI/ML projects on GitHub & Hugging Face, including LLM agents and recommendation systems.
- Took multiple ideas from zero to deployed product without supervision — owning design, build, and release.